

Is this AI?

The two-question test

Almost everything sold as artificial intelligence today is one of three quite different machines. By the end of this session you will be able to tell them apart in under a minute — and you will know what you are going to build between now and December.

Unit 0 • What AI is

80 minutes

No mathematics today

Setup lab + a 30-minute diagnostic

[The quiz](#) [The obvious answer](#) [Two questions](#) [Back to the quiz](#) [Nested boxes](#) [The course](#) [Setup lab](#) [Diagnostic](#) [Before Friday](#)

PART 1 • 15 MINUTES

Six systems. Which of them are artificial intelligence?

Answer for yourself before reading on. Write down yes or no for each. Do not look for a trick — answer the way you would answer a friend.

1

#	System	What it does
A	A thermostat	Turns the air conditioning on when the room passes 26 °C.
B	Deep Blue	Beat the world chess champion in 1997 by examining 200 million positions per second.
C	A spam filter	Learned from a million labelled emails which ones you would have deleted.
D	Google Maps routing	Finds the fastest road from Grand-Bassam to Abidjan through a graph of every street.
E	A pocket calculator	Computes a square root to twelve decimal places, instantly, every time.
F	ChatGPT	Writes an essay on the Ivorian cocoa trade that you would accept from a student.

How does a room usually answer?

Aggregated across several dozen audiences, before any discussion.

Almost everyone says yes to F and no to A and E. Most say yes to B and C. D splits the room roughly in half — and that split is the interesting part of this whole session.

Now hold on to your own answers. We are going to come back and change some of them, on purpose, with a reason.

PART 2 • 15 MINUTES

The answer that sounds right and is not

There is a tempting one-line definition of AI. It is popular, it is repeated in textbooks, and it falls apart on the second example you throw at it.

2

The tempting definition

Ask most people to define artificial intelligence and, after some hedging, you get something close to this:

THE DEFINITION WE ARE ABOUT TO REJECT

“It is AI if a human did not write the rules — if the machine worked them out for itself.”

It is an attractive test. It is crisp, it is easy to apply, and it correctly rejects the thermostat and the calculator. A human wrote *if temperature > 26 then switch on*; a human wrote the square-root algorithm. Neither machine worked anything out. Good so far.

Now apply it to Deep Blue.

Every rule inside Deep Blue was written by a human being. The rules of chess, hand-coded. The evaluation function that scored a position — material, king safety, pawn structure — hand-tuned by grandmasters. The search procedure, hand-written. Deep Blue learned nothing. It did not improve between games. Switch it off and on and it is exactly the machine it was before.

So by the tempting definition, Deep Blue is **not** artificial intelligence. And yet it did the thing that, for forty years, everyone had agreed would be the proof that machines could think: it beat Garry Kasparov.



Where the definition breaks

“Did a human write the rules?” is not a definition of artificial intelligence. It is a definition of *machine learning* — which is one branch of AI, not the whole of it. Treating the two as the same thing quietly deletes forty years of the field, and it will make the rest of this course incoherent.

This is worth stating plainly because it is the single most common confusion in the room, including among people who work with these systems every day. In 2026 the words *AI* and *machine learning* are used interchangeably in press releases, job adverts and, frequently, in conference talks. They are not the same, and you will spend the rest of this semester on the difference.

Two questions, asked in order

One question was never going to be enough. Two, asked in the right sequence, will carry you through the entire field.

3

THE TEST — USE THIS FOR THE REST OF THE SEMESTER

Question 1 — Does the task require intelligence at all?

If a human doing this task would be said to be thinking — perceiving, reasoning, planning, judging, using language — then the task is a candidate. If a human doing it would just be following a fixed procedure, it is not.

Question 2 — Where did the rules come from?

Only ask this if question 1 came back yes. Did a person write the rules down, or did the machine extract them from data? This does not decide whether it is AI. It decides *what kind*.

The order matters. Question 1 is the gate: it separates artificial intelligence from ordinary computation. Question 2 is the classifier: it tells you which of three families you are looking at, and therefore what you should expect from the system, how it fails, and what it would take to improve it.

Question 1, applied

A thermostat compares two numbers. A human doing that is not thinking, they are comparing two numbers. **Not AI** — and notice we never had to ask who wrote the rule.

A calculator executes an algorithm that is exact, known, and finite. A human doing long division is not exercising judgement.

Not AI

Playing chess well, recognising which email is unwanted, finding a good route through a city you do not know, writing coherent prose about the cocoa trade — a human doing any of these would be described as thinking. **All four pass question 1**



Question 1 is a judgement call, and that is acceptable

There is no bright line, and philosophers have argued about it for seventy years. You will meet genuinely borderline cases — is optical character recognition intelligence, or is it pattern matching that stopped being impressive? The test is still useful. A test that gives you a clear answer on ninety per cent of cases and an honest argument on the rest is worth far more than a crisp rule that gives you confidently wrong answers.

Question 2, applied — the three paradigms

Of the systems that passed question 1, ask where the rules came from. You get three distinct families. We will spend all of session 2 on these; today, just meet them.

THREE PARADIGMS — THE SAME FIELD, THREE DIFFERENT MACHINES

RULES-BASED

A human wrote the rules



IF ... THEN ...

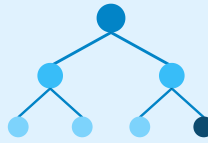
Expert systems

Medical diagnosis, 1980s

Transparent · brittle · caps out
at what the expert could say

SEARCH-BASED

A human wrote the goal



Deep Blue · route finding

Explores possibilities faster
than any human could

LEARNING-BASED

The machine found the rules



weights
adjusted by data

Spam filter · ChatGPT

Everything since 2012

Powerful · opaque · needs
enormous amounts of data

All three are artificial intelligence. All three pass question 1. They differ only in the answer to question 2 — and that difference determines everything practical about them.

PART 4 · 15 MINUTES

Back to the six systems

Now with the test in hand. Compare these against the answers you wrote at the start.

4

Not AI

A · The thermostat · E · The calculator

Both fail at question 1.

Comparing two numbers is not thinking; computing a square root by a known exact algorithm is not thinking. We never reach question 2, and it is worth noticing that we did not need to — asking "did a human write the rule?" first would have given the right answer here for entirely the wrong reason.

A caveat you should be able to make yourself: a modern learning thermostat, which builds a model of when you are home and what you like, *does* pass question 1. The device on the wall in 1970 does not. Same word, different machine.

AI

Search-based

B • Deep Blue • D • Google Maps routing

Pass question 1. On question 2: humans wrote every rule.

These two are the same machine wearing different clothes, and this is why D split the room. Both take a problem with an astronomical number of possible answers, and both search it far faster and far deeper than a person could — Deep Blue through chess positions, Maps through street networks. Neither learned anything. Both are unambiguously AI, and search is one of the oldest and most successful branches of the field.

If you said yes to Deep Blue and no to Google Maps, you were not being inconsistent — you were responding to how *impressive* each one feels. That instinct is worth naming, because it is the source of a long-running joke in this field: as soon as something works reliably, people stop calling it AI. Routing, spell-check, optical character recognition and speech transcription were all once flagship AI results. Now they are just software.

AI

Learning-based

C • The spam filter • F • ChatGPT

Pass question 1. On question 2: the machine extracted the rules from data.

Nobody wrote a rule saying that the word "lottery" plus an unfamiliar sender plus five exclamation marks means spam. The filter found that pattern in a million labelled examples. Nobody wrote a rule about the cocoa trade either. The scale differs by a factor of a billion; the principle is identical.

This third family is what the rest of CSC 4315 is about — not because the other two are unimportant, but because this is the one you can build from scratch in a semester, and the one that has changed since 2012.

THE POINT OF THE WHOLE EXERCISE

The six systems sort into three groups, not two. Anyone who tells you that AI means "the machine learned it" has handed you a test that classifies Deep Blue, Google Maps and every expert system as not-AI — which is a strange thing to say about the programs that carried the field for its first forty years.

Not synonyms — nested boxes

One more piece of vocabulary before we finish, because these five words are used interchangeably in the press and they are not interchangeable at all.

EACH TERM SITS INSIDE THE ONE BEFORE IT

ARTIFICIAL INTELLIGENCE

any machine doing something we would call thinking · rules, search or learning

MACHINE LEARNING

the subset where the rules come from data, not from a person

DEEP LEARNING

machine learning using many-layered neural networks · the 2012 renaissance

LARGE LANGUAGE MODELS

deep learning applied to text at enormous scale · GPT, Claude, Gemini, Llama

an AGENT is an LLM given tools and a loop — a shape, not a layer

Read it outward and each sentence is true: every large language model is deep learning, all deep learning is machine learning, all machine learning is AI. Read it inward and every sentence is false. Deep Blue lives in the outermost box only. Your spam filter lives in the second box but not the third.

PART 5 · 10 MINUTES

What you will build between now and December 5

This course has an unusual rule, and it is the reason it is organised the way it is.

THE RULE OF THIS COURSE

You will not call a library function until you have written the thing it replaces.

By session 14 you will have built a neural network in NumPy — the forward pass by hand on paper, then in code; the gradients derived on the board, then coded, then verified numerically. Only in session 17 will you be allowed to write `model.fit()`. When you do, you will be able to name every line of your own code that the call replaced.

1

Sessions 1–3 · What AI is

Today's test, the three paradigms, eighty years of history including two collapses, and an honest account of where the field actually stands in 2026.

2

Sessions 4–9 · The tools you need and nothing more

Two sessions of Python, two of NumPy, the small amount of linear algebra a network actually uses, and gradient descent worked out with pencil and paper before any code. Six sessions, deliberately unhurried.

3

Sessions 10–14 · A neural network from nothing

No libraries. Least squares, one neuron, the XOR wall, a forward pass you compute by hand, and backpropagation you derive yourself. The heart of the course.

4

Sessions 16–19 · What the libraries were doing all along

Your two hundred lines become four, in scikit-learn, PyTorch and Keras, shown side by side against your own code.

5

Sessions 20–23 · Data that fights back

Missing values, leakage, class imbalance, overfitting, and images at a scale where the tricks matter.

6

Sessions 24–26 · Language, and a model you adapt

Tokens, embeddings, attention computed by hand on three words — and then a small language model, already trained, that you fine-tune on a corpus of your choosing and watch change voice.

7

Sessions 27–28 · Getting real work out of frontier models

Prompting as an engineering discipline, then agents, tools and retrieval. Placed at the end deliberately: this material reads completely differently once you have taken a model apart yourself and put one back to work.



Dates

Thirty sessions, Tuesdays 16:00–17:20 and Fridays 11:30–12:50, from 8 September to 18 December 2026.

Midterm: Tuesday 27 October

, covering sessions 1–14.

Final: Friday 18 December.



On using ChatGPT, Claude or Gemini for your lab work

You may, from session 4 onward, under one condition:

you must be able to explain every line you submit.

I will ask. A student who uses a frontier model to learn faster is doing exactly what a working engineer does in 2026; a student who uses one to avoid understanding will discover the problem at the midterm, where there is no laptop.

Session 27 turns this into a formal skill.

PART 6 · 20 MINUTES · IN CLASS

Setup: get your machine ready

6

Everything in this course runs on a modest laptop. Do this now, in the room, so that problems surface while I am standing here.

1

Install Miniconda

From `docs.conda.io`. Accept the defaults. On Windows, use the Anaconda Prompt for everything below, not PowerShell.

Create the course environment

2 One environment for the whole semester, so that everyone's results are reproducible and my debugging advice applies to your machine.

3 Verify it

Run the check script below. If it prints four version numbers and a green tick, you are done for today.

4 Join the GitHub Classroom

Accept the invitation link. Labs are submitted there — never by email.

Anaconda Prompt (Windows) or Terminal (macOS / Linux)

SHELL

```
# one environment for the whole semester
conda create -n csc4315 python=3.11 -y
conda activate csc4315

# everything we need through session 22
pip install numpy matplotlib scikit-learn jupyterlab pandas

# PyTorch arrives in session 18 — install it now while you have wifi
pip install torch --index-url https://download.pytorch.org/whl/cpu

# launch the notebook server
jupyter lab
```

check_setup.py — run this in a notebook cell

PYTHON

```
import sys, numpy as np, matplotlib, sklearn

print("python      ", sys.version.split()[0])
print("numpy       ", np.__version__)
print("matplotlib   ", matplotlib.__version__)
print("scikit-learn", sklearn.__version__)

# the one line of real work: a 2x3 layer, which is session 7's whole lesson
X = np.array([[1.0, 2.0]])           # one example, two features
W = np.array([[0.5, -1.0, 2.0],
              [1.5, 0.0, -0.5]])     # 2 inputs -> 3 hidden units
b = np.array([0.1, 0.1, 0.1])

out = X @ W + b
print("\nlayer output", out)
assert np.allclose(out, [[3.6, -0.9, 1.1]]), "something is wrong"
print("✅ setup verified")
```



You just ran a neural network layer

Those three lines are not a toy. `X @ W + b` is exactly what happens inside every layer of every network in this course, including the language model you will fine-tune in session 26. Session 7 is devoted to why. For now, notice that you can check the first number by hand: $1.0 \times 0.5 + 2.0 \times 1.5 + 0.1 = 3.6$.

A page of code, and it counts for nothing

The last thirty minutes are a written exercise. Read this paragraph before you start it, because how you read it decides whether it is any use to me.



Not graded. Not for marks, not for participation, not at all.

I am not measuring you. I am measuring the class, so that sessions 4 to 9 get built for the people actually in this room rather than for the students the syllabus imagined. A blank page from someone who has never programmed is a useful answer and I need it. A copied page is worse than useless, because it makes me build the wrong course.

How to answer it

1

Any language.

Python, Java, C, or plain pseudocode. I am reading for whether the idea is right, not whether the semicolons are.

2

On paper.

Closed book, no phones, no laptop — not for integrity, but because an assisted answer tells me nothing about what to teach.

3

Write “not yet” and move on

if you cannot do a question. That is a real answer rather than a failure, and it is the one I most need to count accurately.

4

Put your name on it.

The names are how I pair you sensibly in labs and find the three or four people I should speak to individually this week.

What is on it

Four short parts. A page about your background; three fragments of code to read and say what they print; three small functions to write; and one broken function with three bugs in it to find. There is an optional last page for anyone who already knows NumPy — blank is expected there and costs nothing.



You will hear the result on Friday

In session 2 I will spend ninety seconds on how the class distributed and what I changed because of it. Two things to hold on to until then. Unit 1 is

six sessions

of Python and NumPy — nearly a fifth of the course — precisely because arriving without programming is expected here rather than penalised. And

no lab is a prerequisite for the next lecture

: every session starts from a state the whole room shares, so a week where you fall behind never compounds into the next one.

Before Friday

1. **Finish the setup** if it did not complete in class. Bring me the error message, not a description of it.
2. **Write down one system you use weekly** — an app, a website, a device — and classify it with the two-question test. Bring a one-paragraph justification. We will argue about several of them on Friday, and the disagreements are the useful part.
3. **Optional, ten minutes:** ask a frontier model "is Google Maps routing artificial intelligence?" and see whether its answer survives today's test. Bring what it said.



Session 2 — Friday 11 September, 11:30

Three paradigms in depth: what an expert system actually looked like, why Deep Blue's search was a triumph and a dead end at the same time, and what changed when the third paradigm finally worked.